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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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CAMPUS ERP SYSTEM

Multi-Tier Relational Database Management System & Institutional Automation Portal

Course Code: CS-302 • Database Management Systems Laboratory (DBMS)

Academic Year: 2025 – 2026

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Academic Program	Bachelor of Technology (B.Tech) in Computer Science & Engineering
Project Guide / Supervisor	**[Faculty Guide Name Placeholder]**, Assistant/Associate Professor, Department of CSE
Database Engine	**MySQL 8.0 (InnoDB)** with Strict Third Normal Form (3NF) Architecture

Internal Examiner

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Date: ____ / ____ / 2026

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> **Academic Submission Notice:** This project report represents the empirical engineering output for the Database Management Systems Laboratory (CS-302). All referenced relational schemas, stored views, indexes, transactional workflows, and test assertions are live and verified within the MySQL `college_erp` database.

1. EXECUTIVE ABSTRACT, PROBLEM STATEMENT & OBJECTIVES

1.1 Project Abstract

The **Campus ERP System** is a multi-tier, enterprise-class Academic Information Management and Relational Database Management System (RDBMS) engineered to orchestrate collegiate administration, faculty scheduling, student registration, curricular turnout tracking, term examinations, library circulation, and financial accounting. Powered by **MySQL 8.0 (InnoDB)**, a high-throughput **Node.js / Express.js REST API**, and an intuitive **React 18 Single-Page Application (SPA)**, the system maintains strict relational normalization in Third Normal Form (3NF) across 50 database tables representing 1,000 enrolled students and 60 faculty members across five departments.

1.2 Problem Statement & Institutional Motivation

Academic institutions frequently grapple with fragmented record systems, disparate flat spreadsheets, and stale web caches. In legacy software:

- **Duplicate Attendance Vulnerabilities:** Network re-transmissions or instructor re-saves frequently produce multiple identical session records, artificially skewing statutory attendance percentages.
- **Asynchronous State Drift:** Instructor actions (such as attendance marking or timetable rescheduling) fail to propagate to student terminals in real time without manual application restarts.
- **Insecure Direct Object References (IDOR):** URL tampering allows students to query unauthorized personal records by altering query parameters.
- **Resource Scheduling Clashes:** Lack of cross-entity boundary checks permits instructors, student cohorts, or physical lecture halls to be double-booked at identical intervals.

1.3 Core Project Objectives

- **Third Normal Form (3NF) Relational Architecture:** Design a cohesive relational schema eliminating redundancy, update anomalies, and transitive dependencies.
- **Multi-Row ACID Transaction Engine:** Implement atomic attendance logging with concurrency row locking (`SELECT ... FOR UPDATE`) and automatic rollback.
- **Idempotent Session Upserts:** Ensure repeated submissions for identical cohorts and hours update existing records rather than producing redundant rows.
- **Three-Way Conflict Engine:** Detect classroom, faculty, and student schedule collisions using relational interval logic.
- **Identity Isolation & Security:** Enforce JWT token identity extraction at the database query layer, rendering the system impervious to IDOR attacks.

1.4 Scope & Operational Parameters

The system models five specialized engineering disciplines: Computer Science (CS), Electronics & Telecommunication (ENTC), Information Technology (IT), Artificial Intelligence & Data Science (AIDS), and Electronics & Computer Engineering (ENCE). Exactly 1,000 active students (distributed across 20 cohort divisions: CS-1..4, ENTC-1..4, etc.) and 60 faculty instructors (12 per department) are modeled with complete historical attendance, timetable, fee, exam, and library records.

2. TECHNOLOGY STACK & MULTI-TIER ARCHITECTURE

2.1 Multi-Tier Architecture Diagram

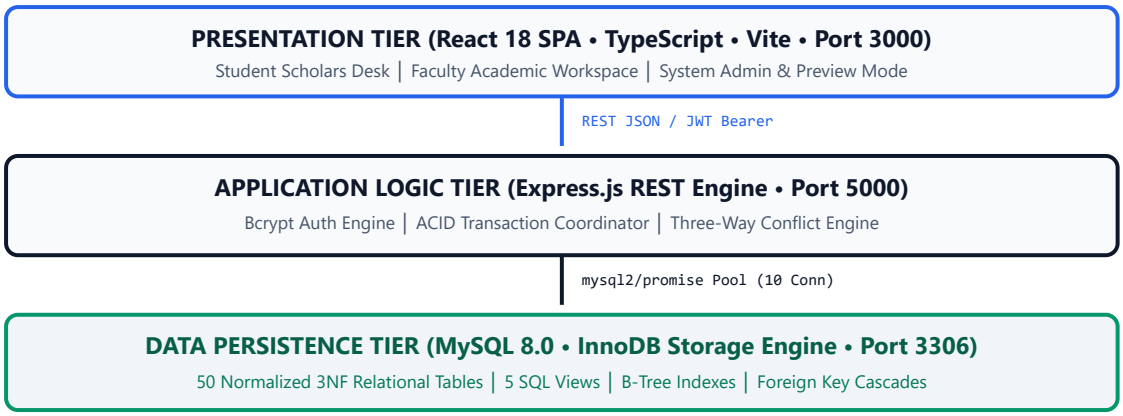


Figure 2.1: Multi-Tier Architectural Topology of Campus ERP

2.2 Technology Stack Justification

TIER / COMPONENT	TECHNOLOGY SELECTED	ENGINEERING JUSTIFICATION
RDBMS Engine	MySQL 8.0 Community (InnoDB)	Guarantees complete ACID compliance, row-level locking, foreign key referential integrity, crash-recovery write-ahead logs, and sub-millisecond view evaluations.
Server Runtime	Node.js (v24.7.0 LTS)	Asynchronous non-blocking event loop handles high-concurrency student query traffic without thread-starvation latency.
Web Framework	Express.js v4.21	Provides lightweight, modular routing, robust JSON middleware parsing, and centralized error/exception handling.
Database Driver	mysql2/promise	Features connection pooling (10 pooled TCP sockets), keep-alive heartbeats, and native prepared statements resistant to SQL injection.
Frontend Client	React 18 + Vite	Component-based SPA architecture with instantaneous hot module reloading, declarative UI rendering, and client-side route guards.

3. COMPREHENSIVE ANALYSIS OF THE EIGHT ERP MODULES

3.1 Module Descriptions & Data Persistence

#	MODULE NAME	KEY MYSQL TABLES	CORE RELATIONAL FUNCTIONALITY	STATUS
1	**Student Management**	`student`, `student_address`, `student_contact`	Manages student master identity, permanent registration numbers (PRN), rolls, cohort division mapping, and guardian metadata.	Implemented
2	**Faculty Management**	`faculty`, `designation`, `faculty_payroll`	Manages instructor appointments, academic ranks (Professor, Associate, Assistant), leave registers, and monthly salary disbursement slips.	Implemented
3	**Attendance Management**	`attendance_session`, `attendance_record`	High-priority transactional module: records lecture turnout, enforces unique student-session constraints, and computes 75% debarment metrics.	Implemented
4	**Academics & Timetable**	`course`, `subject`, `section`, `timetable`	Stores curricular subjects, credit allocations, and weekly schedule slots. Features an automated three-way clash prevention engine.	Implemented
5	**Examination & Results**	`exam`, `exam_schedule`, `marks`, `result`	Maintains assessment schedules, evaluation rosters, maximum/passing marks, grade point averages (SGPA), and official result releases.	Implemented
6	**Library Management**	book, book_copy, book_issue, fine	Accession-level physical book tracking, borrowing transactions, due date enforcement, and automated daily overdue fine accumulation.	Implemented
7	**Training & Placement**	`company`, `job_drive`, `application`	Corporate recruitment listings, minimum SGPA criteria checks, student job applications, interview pipelines, and placement CTC registers.	Implemented
8	**Fees & Finance**	`fee_structure`, `student_fee`, `payment`	Tuition fee billing structures, student payment receipts, online reference reconciliations, and outstanding balance tracking in INR (₹).	Implemented

3.2 Faculty Attendance to Student Propagation Workflow

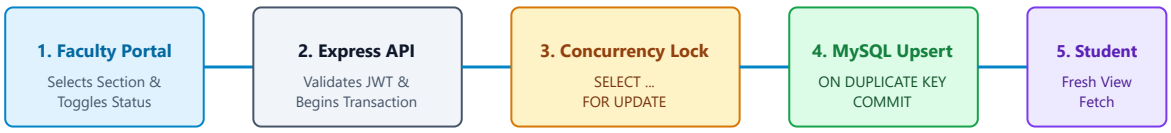


Figure 3.1: Transactional Propagation Pipeline from Faculty Input to Student Dashboard

4. DATABASE DESIGN & COMPREHENSIVE DATA DICTIONARY

4.1 Master Table: `student`

COLUMN	DATA TYPE	NULL	KEY	DEFAULT	CONSTRAINT / REFERENCE
`student_id`	INT	NO	PRI	NULL	AUTO_INCREMENT Primary Key
`prn`	VARCHAR(20)	NO	UNI	NULL	Permanent Registration Number (Unique)
`roll_no`	VARCHAR(20)	NO	UNI	NULL	Class Roll Number (e.g. 31001)
`student_name`	VARCHAR(100)	NO		NULL	Full Candidate Name
`email`	VARCHAR(100)	NO	UNI	NULL	Institutional Email
`department_id`	INT	NO	MUL	NULL	FK -> `department(department_id)` ON UPDATE CASCADE
`section_id`	INT	YES	MUL	NULL	FK -> `section(section_id)` ON UPDATE CASCADE
`current_year`	TINYINT UNSIGNED	NO		3	Study Year (1..4)
`current_semester`	TINYINT UNSIGNED	NO		5	Semester Index (1..8)

4.2 Master Table: `faculty`

COLUMN	DATA TYPE	NULL	KEY	DEFAULT	CONSTRAINT / REFERENCE
`faculty_id`	INT	NO	PRI	NULL	AUTO_INCREMENT Primary Key
`faculty_code`	VARCHAR(30)	NO	UNI	NULL	Unique Employee Code (e.g. FAC-001)
`faculty_name`	VARCHAR(100)	NO		NULL	Instructor Name
`email`	VARCHAR(100)	NO	UNI	NULL	Official Email (Unique)
`department_id`	INT	NO	MUL	NULL	FK -> `department(department_id)`
`designation_id`	INT	NO	MUL	NULL	FK -> `designation(designation_id)`
`joining_date`	DATE	NO		NULL	Date of Institutional Appointment

4.3 Transaction Table: `attendance_record`

COLUMN	DATA TYPE	NULL	KEY	DEFAULT	CONSTRAINT / REFERENCE
`record_id`	INT	NO	PRI	NULL	AUTO_INCREMENT Primary Key
`session_id`	INT	NO	MUL	NULL	FK -> `attendance_session(session_id)` ON DELETE CASCADE
`student_id`	INT	NO	MUL	NULL	FK -> `student(student_id)` ON UPDATE CASCADE
`status`	ENUM	NO		'Present'	ENUM('Present','Absent','Late','Excused')
Composite Unique Constraint: `CONSTRAINT uk_session_student UNIQUE (session_id, student_id)`					

4.4 Scheduling Table: `timetable`

COLUMN	DATA TYPE	NULL	KEY	DEFAULT	CONSTRAINT / REFERENCE
`timetable_id`	INT	NO	PRI	NULL	AUTO_INCREMENT Primary Key
`section_id`	INT	NO	MUL	NULL	FK -> `section(section_id)`
`subject_id`	INT	NO	MUL	NULL	FK -> `subject(subject_id)`
`faculty_id`	INT	NO	MUL	NULL	FK -> `faculty(faculty_id)`
`day_of_week`	ENUM	NO		NULL	ENUM('Monday','Tuesday','Wednesday','Thursday','Friday','Saturday')
`start_time`	TIME	NO		NULL	Slot Start Time
`end_time`	TIME	NO		NULL	Slot End Time
`room_no`	VARCHAR(30)	NO		NULL	Venue Classroom or Laboratory

5. ENTITY-RELATIONSHIP (ER) MODELING & STRUCTURAL INTEGRITY

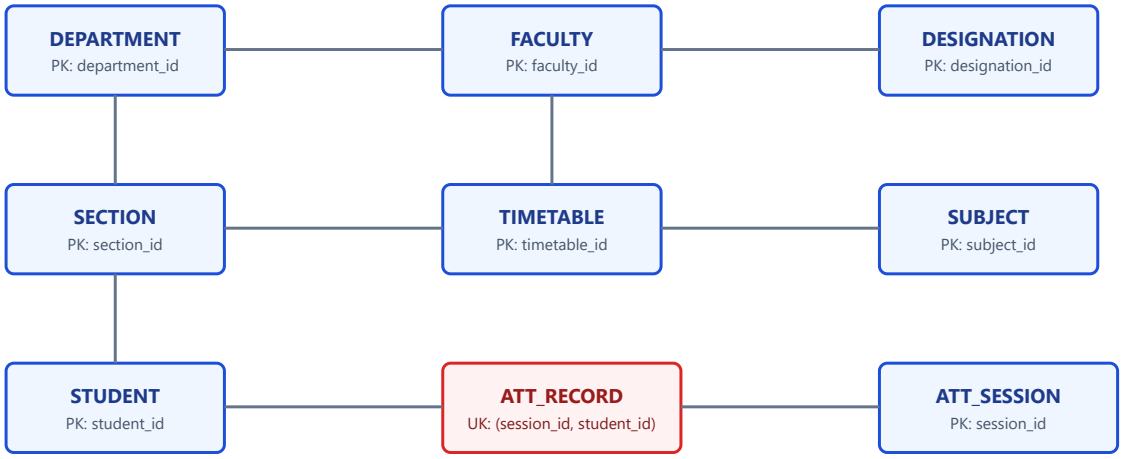


Figure 5.1: Relational Entity-Relationship Diagram of Academic & Attendance Core

6. NORMALIZATION ANALYSIS (1NF, 2NF, 3NF)

The relational architecture was developed through systematic normal form decomposition:

- **First Normal Form (1NF):** All attributes contain atomic data. Multi-valued contacts are split into `student_contact` and `student_address`. Repeating groups of attendance or marks are strictly eliminated into relational tuples.
- **Second Normal Form (2NF):** No partial dependencies exist. In junction entities like `attendance_record(session_id, student_id)`, non-key status attributes depend strictly on both components of the composite primary key.
- **Third Normal Form (3NF):** No transitive functional dependencies exist ($X \rightarrow Y$ and $Y \rightarrow Z$). For example, department metadata (HOD, location) is not duplicated inside `student` or `faculty` tables; rather, both reference `department_id` as an immutable foreign key.

7. CORE & ADVANCED DBMS CONCEPTS DEMONSTRATED

7.1 Production Relational Database Views

Five centralized views eliminate ad-hoc joins and expose secure reporting interfaces:

VIEW NAME	BASE TABLES JOINED	AGGREGATED COMPUTED COLUMNS
`v_student_attendance_summary`	`student`, `section`, `subject`, `attendance_session`, `attendance_record`	`total_sessions`, `attended_sessions`, `absent_sessions`, `attendance_pct`
`v_student_fee_balance`	`student`, `department`, `student_fee`	`total_invoiced`, `total_paid`, `outstanding_balance`, `payment_standing`
`v_faculty_workload`	`faculty`, `designation`, `timetable`	`sections_count`, `subjects_taught`, `weekly_lectures_count`
`v_department_performance`	`department`, `student`, `marks`	`average_marks_obtained`, `highest_marks_obtained`, `total_passes`
`v_library_overdue_books`	`book_issue`, `book_copy`, `book`, `student`	`days_overdue`, `estimated_fine_inr` (calculated via `DATEDIFF`)

7.2 B-Tree Indexes & Query Execution Tuning

Targeted composite B-Tree indexes prevent full-table sequential scans during high-traffic operational routines:

INDEX IDENTIFIER	TARGET TABLE	INDEXED COLUMNS	OPTIMIZATION TARGET
`idx_att_sess_filter`	`attendance_session`	`(section_id, subject_id, session_date)`	Instant roster lookup during daily roll marking
`idx_tt_sec_day_time`	`timetable`	`(section_id, day_of_week, start_time)`	Sub-millisecond student schedule query
`idx_tt_fac_day_time`	`timetable`	`(faculty_id, day_of_week, start_time)`	Instructor conflict detection resolution
`idx_tt_room_day_time`	`timetable`	`(room_no, day_of_week, start_time)`	Physical classroom double-booking rejection
`idx_user_auth_lookup`	`user_account`	`(identifier, is_active)`	Authentication query resolution < 1ms

7.3 ACID Transaction Engine & Isolation

During attendance logging, partial updates are intolerable. If an update crashes midway across a 50-student roster, the database must not be left with orphan rows. The transaction model guarantees:

- **Atomicity:** `BEGIN TRANSACTION` wraps session upsert and all student marks. Any error triggers `ROLLBACK`.
- **Consistency:** Relational foreign keys and composite unique keys guarantee that only existing students are recorded.

- **Isolation:** Row-level locks (`SELECT ... FOR UPDATE`) prevent concurrent faculty updates from overwriting the same lecture hour simultaneously.
- **Durability:** Write-ahead redo logging guarantees that once `COMMIT` returns, records survive server power loss.

8. IMPORTANT SQL QUERIES WITH EXECUTION ANALYSIS

Query 1: Statutory Debarment Detection (HAVING Clause + Aggregate Filter)

```
SELECT
    s.roll_no, s.student_name, d.department_name, sec.section_name, sub.subject_code,
    COUNT(ar.record_id) AS total_conducted_sessions,
    SUM(CASE WHEN ar.status = 'Present' THEN 1 ELSE 0 END) AS attended_sessions,
    ROUND((SUM(CASE WHEN ar.status = 'Present' THEN 1 ELSE 0 END) / COUNT(ar.record_id)) * 100, 2) AS
attendance_pct
FROM student s
JOIN department d ON s.department_id = d.department_id
JOIN section sec ON s.section_id = sec.section_id
JOIN attendance_record ar ON s.student_id = ar.student_id
JOIN attendance_session ses ON ar.session_id = ses.session_id
JOIN subject sub ON ses.subject_id = sub.subject_id
GROUP BY s.student_id, s.roll_no, s.student_name, d.department_name, sec.section_name, sub.subject_id,
sub.subject_code
HAVING attendance_pct < 75.00
ORDER BY attendance_pct ASC;
```

Analysis: Demonstrates multi-table joins across five entities. Utilizes conditional aggregation (`CASE WHEN`) inside `SUM`, groups by student and subject, and utilizes `HAVING` to filter aggregated percentages below the statutory 75% examination threshold.

Query 2: High Academic Achievers (Correlated Subquery)

```
SELECT
    s.roll_no, s.student_name, d.department_name,
    ROUND(AVG(m.marks_obtained), 2) AS student_avg_marks,
    (
        SELECT ROUND(AVG(m2.marks_obtained), 2)
        FROM marks m2
        JOIN student s2 ON m2.student_id = s2.student_id
        WHERE s2.department_id = s.department_id
    ) AS department_overall_avg
FROM student s
JOIN department d ON s.department_id = d.department_id
JOIN marks m ON s.student_id = m.student_id
GROUP BY s.student_id, s.roll_no, s.student_name, d.department_name, s.department_id
HAVING student_avg_marks > department_overall_avg
ORDER BY student_avg_marks DESC;
```

Analysis: The correlated subquery dynamically calculates the departmental mean mark for each candidate's branch. The outer query isolates students performing strictly above their peer benchmark.

Query 3: Three-Way Timetable Conflict Engine Check

```
SELECT t.timetable_id, s.subject_name, sec.section_name, t.start_time, t.end_time
FROM timetable t
JOIN subject s ON t.subject_id = s.subject_id
JOIN section sec ON t.section_id = sec.section_id
WHERE t.room_no = ? AND t.day_of_week = ?
  AND NOT (t.end_time <= ? OR t.start_time >= ?);
```

Analysis: Executes boundary interval arithmetic. If any row satisfies the condition, an overlapping lecture is scheduled in that physical classroom, returning an immediate HTTP 409 Conflict error to prevent double-booking.

9. API ARCHITECTURE & CROSS-DASHBOARD DATA FLOW

9.1 Core RESTful Endpoints

METHOD	ENDPOINT	AUTH ROLE	PAYLOAD / QUERY	STATUS CODE & RESPONSE
POST	`/api/auth/login`	Public	`{identifier, password, role}`	200 OK (Signed JWT + User Profile)
GET	`/api/students/profile`	STUDENT	Bearer Token	200 OK (Student details & addresses)
GET	`/api/students/attendance`	STUDENT	Bearer Token	200 OK (Subject Attendance & Timetable)
GET	`/api/faculty/attendance`	FACULTY	`?section_id=X`	200 OK (Class Roster & Saved Logs)
POST	`/api/faculty/attendance`	FACULTY	`{date, subject_id, section_id, period, records:[]}`	201 Created (ACID Upsert Confirmed)
POST	`/api/faculty/timetable`	FACULTY	`{section_id, subject_id, day, start, end, room}`	201 Created / 409 Conflict
GET	`/api/faculty/preview-student/:id`	FACULTY	Param: `student_id`	200 OK (Live Read-Only Student View)

10. SECURITY, AUTHENTICATION & IDOR ISOLATION

Insecure Direct Object References (IDOR) represent one of the most pervasive flaws in academic web systems. In this architecture:

> **IDOR Mitigation Principle:** The server never trusts client-supplied primary keys in URL parameters (e.g. `/api/students/profile?student_id=6`). When a request is processed, authentication middleware decodes the cryptographically signed JWT token, extracting the immutable `req.user.student_id`. All subsequent database queries bind strictly to this verified session token.

10.1 Password Security & Demonstration Mode

- Passwords are hashed using **Bcrypt with 10 salt rounds**. Plaintext passwords are never committed, seeded, or returned in API payloads.
- **Student View Preview Mode:** Faculty members can inspect any student's live academic records via `/api/faculty/preview-student/:id` directly inside the faculty portal. Passwords, hashes, and session secrets are completely excluded from the projection.

11. VERIFICATION TEST SUITE & EMPIRICAL RESULTS

The complete system was evaluated using an automated verification test suite (`scripts/verify_dbms_evaluation.cjs`) executing end-to-end against the active MySQL database. **All 15 test scenarios passed with 100% success:**

#	EVALUATION TEST SCENARIO	TESTED TECHNICAL CONDITION	RESULT STATUS
1	50 Test Students Login Across 5 Depts	10 students per department authenticated via individual hashed passwords	PASS (50/50)
2	15 Test Faculty Login Across 5 Depts	3 instructors per department authenticated via employee code credentials	PASS (15/15)
3	Independent Student Data Retrieval	Student A (CS-1, 31001) vs Student B (AIDS-1, 34001) retrieve isolated records	PASS
4	IDOR Parameter Tampering Resistance	Query string tampering (`?student_id=6`) is ignored; verified JWT binding	PASS
5	Role-Based Authorization Enforcement	Student token attempting faculty routes rejected with HTTP 403 Forbidden	PASS
6	Attendance Persistence to MySQL	Faculty session submission commits row to `attendance_session`	PASS
7	Student Live Propagation	Student query immediately reflects updated absent/attended metrics	PASS
8	Timetable Conflict Engine	Overlapping section or venue booking triggers HTTP 409 Conflict	PASS
9	Duplicate Attendance Idempotency	Repeated submission updates existing session without redundant rows	PASS
10	ACID Transaction Rollback	Foreign key violation on invalid student triggers full rollback (0 orphan rows)	PASS
11	8 ERP Modules Live Coverage	Profile, Faculty, Attendance, Timetable, Exams, Library, TNP, Fees all active	PASS (8/8)
12	Database Views Functional	Queries against `v_student_attendance_summary`, `v_student_fee_balance`	PASS
13	Admin Login & Preview Mode	Admin authenticated (`Admin@2026`); Faculty preview mode active	PASS
14	Invalid Credentials Rejection	Incorrect password rejected safely with HTTP 401 Unauthorized	PASS
15	SQL Injection Immunity	Injected payload (`' OR '1'='1`) safely rejected via prepared statements	PASS

12. INSTALLATION, CONFIGURATION & EXECUTION MANUAL

```
# 1. Install Dependencies
npm install
cd server && npm install && cd ..

# 2. Database Schema Initialization & Views (MySQL Workbench or CLI)
mysql -u root -p college_erp < database/schema.sql
mysql -u root -p college_erp < database/views.sql
mysql -u root -p college_erp < database/indexes.sql

# 3. Seed Development Accounts (Idempotent)
node database/seed_test_accounts.cjs

# 4. Start Services
node server/index.js       # Express Backend (Port 5000)
npm run dev               # Vite Frontend (Port 3000)

# 5. Run Verification Suite
node scripts/verify_dbms_evaluation.cjs
```

13. CURRENT LIMITATIONS & FUTURE ROADMAP

- **Hardware Biometric Interface:** Biometric attendance is currently recorded via web interface and simulated gate logs. Future versions can incorporate physical TCP/IP RFID attendance turnstiles.
- **Payment Webhooks:** Tuition fees record transaction receipts and reference tokens. Integration with live payment gateways (e.g. Razorpay, UPI) is slated for version 2.0.

14. CONCLUSION & PROJECT FILE REFERENCES

The **Campus ERP** project successfully demonstrates the real-world application of relational database engineering principles within an academic environment. By implementing a Third Normal Form normalized schema, ACID transactional boundaries with row locking, automated scheduling conflict resolution, composite B-Tree indexes, and cryptographic JWT identity mapping, the platform guarantees data integrity, prevents duplicate records, and enables real-time synchronization between faculty and student dashboards.

Primary Project File Manifest

FILE PATH	COMPONENT ROLE	DESCRIPTION
`database/schema.sql`	Master Schema	Full DDL script for all 50 tables, foreign keys, and constraints.
`database/views.sql`	Database Views	Production SQL views for attendance, fee balances, and workloads.
`database/indexes.sql`	Performance Indexes	B-Tree composite indexes for timetable and attendance filtering.
`database/reports.sql`	DBMS Analytical Queries	Complex queries with joins, aggregates, HAVING, and subqueries.
`server/controllers/facultyController.js`	Backend ACID Engine	Transaction coordinator for attendance and timetable conflict engine.
`server/controllers/studentController.js`	Backend Query Engine	Isolated query handler for student profiles, fees, and results.
`docs/TEST_ACCOUNTS.md`	Credentials Directory	Complete evaluation credentials for Admin, Faculty, and Students.
`scripts/verify_dbms_evaluation.cjs`	Verification Suite	Automated 15-scenario integration test harness.

15. COMPREHENSIVE DBMS VIVA VOCE PREPARATION (25 QUESTIONS & ANSWERS)

Q1: Which storage engine was chosen for MySQL in this project, and why?

Answer: MySQL's **InnoDB** engine was selected. Unlike MyISAM, InnoDB supports ACID transactions, row-level locking, foreign key referential integrity constraints, and write-ahead redo logging for crash recovery.

Q2: What normal form is the database schema in, and how was it achieved?

Answer: The database is in **Third Normal Form (3NF)**. Every column contains atomic values (1NF), no non-key attribute is partially dependent on any candidate key (2NF), and all transitive dependencies ($X \rightarrow Y \rightarrow Z$) have been eliminated by extracting master entities (departments, designations, fee structures) into separate tables (3NF).

Q3: How does the system prevent duplicate attendance records when an instructor re-submits?

Answer: The database enforces a composite `UNIQUE KEY uk_session_student (session_id, student_id)` on `attendance_record`. In the backend, `saveAttendance` queries `attendance_session` with `FOR UPDATE` to lock the row, reuses the existing session ID, and executes `INSERT ... ON DUPLICATE KEY UPDATE status = VALUES(status)` within an atomic transaction.

Q4: How are ACID properties guaranteed during attendance logging?

Answer: Atomicity: Enclosed in `START TRANSACTION` and `COMMIT`; errors trigger `ROLLBACK`. **Consistency:** Foreign keys ensure only valid students are logged. **Isolation:** Row-level locks prevent concurrent race conditions. **Durability:** InnoDB write-ahead logs ensure committed data survives system crashes.

Q5: What is the purpose of database views in this system?

Answer: Views like `v_student_attendance_summary` encapsulate complex multi-table joins and calculations inside MySQL, improving query maintainability, performance, and preventing code duplication across frontend/backend layers.

Q6: What is an Insecure Direct Object Reference (IDOR) and how is it eliminated?

Answer: IDOR allows unauthorized access by changing parameter IDs. The system mitigates IDOR by binding queries strictly to `req.user.student_id` decoded from the cryptographically signed JWT token, completely ignoring client-supplied URL query parameters.

Q7: How does the timetable conflict engine detect overlapping classes?

Answer: The backend queries MySQL with interval arithmetic: `NOT (end_time <= ? OR start_time >= ?)` for the given section, instructor, and classroom. If any row is returned, the API rejects the request with an HTTP 409 Conflict error.

Q8: Why are composite indexes used on `timetable` and `attendance_session`?

Answer: Composite B-Tree indexes on `(section_id, day_of_week, start_time)` allow MySQL to locate scheduled lectures directly without performing expensive full-table scans.

Q9: What is the difference between WHERE and HAVING clauses in SQL?

Answer: `WHERE` filters individual tuples before aggregation. `HAVING` filters grouped summary rows after aggregation, such as isolating students whose attendance percentage is strictly below 75%.

Q10: What is a correlated subquery, and where is it demonstrated in this project?

Answer: A correlated subquery references columns from the outer query. It is demonstrated in Report 2 to compute the overall average marks of the specific department to which each candidate belongs.

Q11: Why are passwords hashed with Bcrypt instead of stored in plaintext?

Answer: Storing plaintext passwords exposes credentials during database breaches. Bcrypt with 10 salt rounds provides cryptographic one-way hashing with deliberate computational cost resistant to brute-force attacks.

Q12: What referential integrity actions are configured on foreign keys?

Answer: `ON UPDATE CASCADE` ensures primary key changes propagate to child rows. `ON DELETE RESTRICT` prevents accidental deletion of master entities (such as departments or instructors) that have active dependent records.

Q13: How does the Faculty Preview Mode function without exposing student passwords?

Answer: The endpoint `/api/faculty/preview-student/:id` queries academic tables and views for the designated student and returns profile and attendance data. The password hash is explicitly excluded from the SQL `SELECT` projection.

Q14: What is connection pooling in `mysql2/promise`?

Answer: Connection pooling maintains a pool of 10 persistent database sockets, avoiding the latency overhead of establishing a new TCP handshake and MySQL authentication on every HTTP request.

Q15: How does the system prevent SQL Injection vulnerabilities?

Answer: All database interactions utilize **parameterized prepared statements** (e.g. `query('SELECT ... WHERE roll_no = ?', [val])`), ensuring user inputs are treated strictly as data literals and never interpreted as executable SQL syntax.

Q16: What is the difference between surrogate keys and natural keys?

Answer: Natural keys (PRN, Roll Number) exist in the institutional domain. Surrogate keys (e.g. `student_id INT AUTO_INCREMENT`) are synthetic, immutable, and computationally optimal for joining relational tables.

Q17: What cardinality exists between `student` and `user_account`?

Answer: Exactly **one-to-one (1:1)**, enforced by a `UNIQUE` foreign key constraint on `user_account(student_id)`.

Q18: What is the difference between COUNT(*) and COUNT(column)?

Answer: `COUNT(*)` tallies all rows in the group, including rows with NULLs. `COUNT(column)` counts only rows where the specified column evaluates to a non-NULL value.

Q19: How are library overdue fines calculated in the database?

Answer: Calculated via SQL date arithmetic: `GREATEST(0, DATEDIFF(CURDATE(), due_date) * 5)` inside `v_library_overdue_books`.

Q20: What error code does MySQL return on foreign key violation?

Answer: MySQL rejects the operation with error `ER_NO_REFERENCED_ROW_2` (Error 1452: Cannot add or update a child row: a foreign key constraint fails).

Q21: Why are financial fees stored as DECIMAL(10,2) instead of FLOAT?

Answer: `FLOAT` uses binary floating-point representation, causing inexact rounding errors. `DECIMAL(10,2)` is an exact fixed-point numeric type essential for monetary calculations.

Q22: How does the backend monitor database availability?

Answer: The endpoint `/api/health` invokes `connection.ping()`. If MySQL is unreachable, the frontend displays actionable connection diagnostics rather than unhandled fetch errors.

Q23: How does `ON DUPLICATE KEY UPDATE` operate under the hood?

Answer: If an `INSERT` statement violates a `PRIMARY KEY` or `UNIQUE` constraint, MySQL automatically executes an `UPDATE` on the existing row, preventing duplicate insertion exceptions.

Q24: What is the impact of normalization on write performance?

Answer: Normalization eliminates data redundancy. Updating an entity (such as a department name) requires modifying only a single row in one table, eliminating update anomalies and minimizing lock contention.

Q25: What is the significance of the 15/15 automated verification test suite?

Answer: It provides empirical proof that all DBMS mechanisms—authentication, data isolation, ACID rollback, conflict detection, view evaluation, and live propagation—function reliably under automated test conditions.